

# Predicting House Prices

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## House Price Prediction

### Introduction

This project participates in the **Kaggle House Prices - Advanced Regression Techniques** competition. The goal is to predict the final sale price of residential homes in Ames, Iowa, using a comprehensive dataset containing 79 explanatory variables describing every aspect of the houses (from square footage and quality to features like fireplaces, garages, and pool quality).

Successful prediction of house prices is a classic regression problem with significant practical value in real estate analytics. This notebook demonstrates a complete end-to-end data science workflow — including extensive exploratory data analysis, feature engineering, handling of missing values, non-linear relationships, multicollinearity, and model diagnostics — culminating in a high-performing linear regression model.

### Project Objectives

- Achieve a strong predictive performance on unseen test data (measured by Root Mean Squared Logarithmic Error - RMSLE)
- Build a well-specified and interpretable linear regression model
- Thoroughly validate model assumptions and document the modeling decisions
- Submit competitive predictions to the Kaggle leaderboard

The final model achieves an **Adjusted R<sup>2</sup> of 90.16%** on the training data and satisfies the core assumptions of linear regression to a high standard.

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It's crucial to get an idea of the data set and a summary statistic is often useful to gain information on our data. We first read in the data and mutate our character variables to factors for analysis purposes when we need them. For now we will not as the following analysis is going to only consider the numerical variables.

```
train <- read.csv("train.csv")

library(dplyr)
train <- train %>%
  mutate(across(where(is.character), as.factor))

train$MSSubClass <- as.factor(train$MSSubClass)
summary(train)
```

```

##      Id      MSSubClass      MSZoning      LotFrontage
## Min.    : 1.0    20      :536    C (all): 10    Min.    : 21.00
## 1st Qu.: 365.8  60      :299    FV      : 65    1st Qu.: 59.00
## Median : 730.5  50      :144    RH      : 16    Median : 69.00
## Mean   : 730.5  120     : 87    RL      :1151   Mean   : 70.05
## 3rd Qu.:1095.2  30      : 69    RM      : 218   3rd Qu.: 80.00
## Max.   :1460.0  160     : 63           Max.   :313.00
##              (Other):262           NAs     :259
##      LotArea      Street      Alley      LotShape      LandContour      Utilities
## Min.    : 1300    Grvl: 6    Grvl: 50    IR1:484    Bnk: 63    AllPub:1459
## 1st Qu.: 7554    Pave:1454    Pave: 41    IR2: 41    HLS: 50    NoSeWa: 1
## Median : 9478           NAs :1369    IR3: 10    Low: 36
## Mean   : 10517           Reg:925    Lvl:1311
## 3rd Qu.: 11602
## Max.   :215245
##
##      LotConfig      LandSlope      Neighborhood      Condition1      Condition2
## Corner : 263      Gtl:1382      NNames :225      Norm :1260      Norm :1445
## CulDSac: 94      Mod: 65      CollgCr:150      Feedr : 81      Feedr : 6
## FR2 : 47      Sev: 13      OldTown:113      Artery : 48      Artery : 2
## FR3 : 4           Edwards:100      RRAn : 26      PosN : 2
## Inside :1052      Somerst: 86      PosN : 19      RRNn : 2
##              Gilbert: 79      RRAe : 11      PosA : 1
##              (Other):707      (Other): 15      (Other): 2
##      BldgType      HouseStyle      OverallQual      OverallCond      YearBuilt
## 1Fam :1220      1Story :726      Min. : 1.000      Min. :1.000      Min. :1872
## 2fmCon: 31      2Story :445      1st Qu.: 5.000      1st Qu.:5.000      1st Qu.:1954
## Duplex: 52      1.5Fin :154      Median : 6.000      Median :5.000      Median :1973
## Twnhs : 43      SLvl : 65      Mean : 6.099      Mean :5.575      Mean :1971
## TwnhsE: 114      SFoyer : 37      3rd Qu.: 7.000      3rd Qu.:6.000      3rd Qu.:2000
##              1.5Unf : 14      Max. :10.000      Max. :9.000      Max. :2010
##              (Other): 19
##      YearRemodAdd      RoofStyle      RoofMatl      Exterior1st      Exterior2nd
## Min. :1950      Flat : 13      CompShg:1434      VinylSd:515      VinylSd:504
## 1st Qu.:1967      Gable :1141      Tar&Grv: 11      HdBoard:222      MetalSd:214
## Median :1994      Gambrel: 11      WdShngl: 6      MetalSd:220      HdBoard:207
## Mean :1985      Hip : 286      WdShake: 5      Wd Sdng:206      Wd Sdng:197
## 3rd Qu.:2004      Mansard: 7      ClyTile: 1      Plywood:108      Plywood:142
## Max. :2010      Shed : 2      Membran: 1      CemntBd: 61      CmentBd: 60
##              (Other): 2      (Other):128      (Other):136
##      MasVnrType      MasVnrArea      ExterQual      ExterCond      Foundation      BsmtQual
## BrkCmn : 15      Min. : 0.0      Ex: 52      Ex: 3      BrkTil:146      Ex :121
## BrkFace:445      1st Qu.: 0.0      Fa: 14      Fa: 28      CBlock:634      Fa : 35
## None :864      Median : 0.0      Gd:488      Gd: 146      PConc :647      Gd :618
## Stone :128      Mean : 103.7      TA:906      Po: 1      Slab : 24      TA :649
## NAs : 8      3rd Qu.: 166.0      TA:1282      Stone : 6      NAs: 37
##              Max. :1600.0      Wood : 3
##              NAs :8
##      BsmtCond      BsmtExposure      BsmtFinType1      BsmtFinSF1      BsmtFinType2
## Fa : 45      Av :221      ALQ:220      Min. : 0.0      ALQ: 19
## Gd : 65      Gd :134      BLQ:148      1st Qu.: 0.0      BLQ: 33
## Po : 2      Mn :114      GLQ:418      Median : 383.5      GLQ: 14
## TA :1311      No :953      LwQ: 74      Mean : 443.6      LwQ: 46
## NAs: 37      NAs: 38      Rec:133      3rd Qu.: 712.2      Rec: 54

```

```

##                               Unf:430           Max.   :5644.0   Unf:1256
##                               NAs: 37             NAs: 38
##      BsmtFinSF2      BsmtUnfSF      TotalBsmtSF      Heating      HeatingQC
## Min.   : 0.00   Min.   : 0.0   Min.   : 0.0   Floor:   1   Ex:741
## 1st Qu.: 0.00   1st Qu.: 223.0   1st Qu.: 795.8   GasA :1428   Fa: 49
## Median : 0.00   Median : 477.5   Median : 991.5   GasW : 18   Gd:241
## Mean   : 46.55   Mean   : 567.2   Mean   :1057.4   Grav : 7    Po: 1
## 3rd Qu.: 0.00   3rd Qu.: 808.0   3rd Qu.:1298.2   OthW : 2    TA:428
## Max.   :1474.00   Max.   :2336.0   Max.   :6110.0   Wall : 4
##
##      CentralAir Electrical      X1stFlrSF      X2ndFlrSF      LowQualFinSF
## N: 95      FuseA: 94   Min.   : 334   Min.   : 0   Min.   : 0.000
## Y:1365      FuseF: 27   1st Qu.: 882   1st Qu.: 0   1st Qu.: 0.000
##              FuseP: 3   Median :1087   Median : 0   Median : 0.000
##              Mix : 1   Mean   :1163   Mean   : 347   Mean   : 5.845
##              SBrkr:1334   3rd Qu.:1391   3rd Qu.: 728   3rd Qu.: 0.000
##              NAs : 1   Max.   :4692   Max.   :2065   Max.   :572.000
##
##      GrLivArea      BsmtFullBath      BsmtHalfBath      FullBath
## Min.   : 334   Min.   :0.0000   Min.   :0.00000   Min.   :0.000
## 1st Qu.:1130   1st Qu.:0.0000   1st Qu.:0.00000   1st Qu.:1.000
## Median :1464   Median :0.0000   Median :0.00000   Median :2.000
## Mean   :1515   Mean   :0.4253   Mean   :0.05753   Mean   :1.565
## 3rd Qu.:1777   3rd Qu.:1.0000   3rd Qu.:0.00000   3rd Qu.:2.000
## Max.   :5642   Max.   :3.0000   Max.   :2.00000   Max.   :3.000
##
##      HalfBath      BedroomAbvGr      KitchenAbvGr      KitchenQual      TotRmsAbvGrd
## Min.   :0.0000   Min.   :0.000   Min.   :0.000   Ex:100   Min.   : 2.000
## 1st Qu.:0.0000   1st Qu.:2.000   1st Qu.:1.000   Fa: 39   1st Qu.: 5.000
## Median :0.0000   Median :3.000   Median :1.000   Gd:586   Median : 6.000
## Mean   :0.3829   Mean   :2.866   Mean   :1.047   TA:735   Mean   : 6.518
## 3rd Qu.:1.0000   3rd Qu.:3.000   3rd Qu.:1.000   3rd Qu.: 7.000
## Max.   :2.0000   Max.   :8.000   Max.   :3.000   Max.   :14.000
##
##      Functional      Fireplaces      FireplaceQu      GarageType      GarageYrBlt
## Maj1: 14   Min.   :0.000   Ex : 24   2Types : 6   Min.   :1900
## Maj2: 5    1st Qu.:0.000   Fa : 33   Attchd :870   1st Qu.:1961
## Min1: 31   Median :1.000   Gd :380   Basement: 19   Median :1980
## Min2: 34   Mean   :0.613   Po : 20   BuiltIn: 88   Mean   :1979
## Mod : 15   3rd Qu.:1.000   TA :313   CarPort: 9    3rd Qu.:2002
## Sev : 1    Max.   :3.000   NAs:690   Detchd :387   Max.   :2010
## Typ :1360                               NAs : 81   NAs :81
##      GarageFinish      GarageCars      GarageArea      GarageQual      GarageCond      PavedDrive
## Fin:352   Min.   :0.000   Min.   : 0.0   Ex : 3   Ex : 2   N: 90
## RFn:422   1st Qu.:1.000   1st Qu.: 334.5   Fa : 48   Fa : 35   P: 30
## Unf:605   Median :2.000   Median : 480.0   Gd : 14   Gd : 9    Y:1340
## NAs: 81   Mean   :1.767   Mean   : 473.0   Po : 3    Po : 7
##              3rd Qu.:2.000   3rd Qu.: 576.0   TA :1311   TA :1326
##              Max.   :4.000   Max.   :1418.0   NAs: 81   NAs: 81
##
##      WoodDeckSF      OpenPorchSF      EnclosedPorch      X3SsnPorch
## Min.   : 0.00   Min.   : 0.00   Min.   : 0.00   Min.   : 0.00
## 1st Qu.: 0.00   1st Qu.: 0.00   1st Qu.: 0.00   1st Qu.: 0.00
## Median : 0.00   Median : 25.00   Median : 0.00   Median : 0.00

```

```
## Mean : 94.24 Mean : 46.66 Mean : 21.95 Mean : 3.41
## 3rd Qu.:168.00 3rd Qu.: 68.00 3rd Qu.: 0.00 3rd Qu.: 0.00
## Max. :857.00 Max. :547.00 Max. :552.00 Max. :508.00
##
## ScreenPorch PoolArea PoolQC Fence MiscFeature
## Min. : 0.00 Min. : 0.000 Ex : 2 GdPrv: 59 Gar2: 2
## 1st Qu.: 0.00 1st Qu.: 0.000 Fa : 2 GdWo : 54 Othr: 2
## Median : 0.00 Median : 0.000 Gd : 3 MnPrv: 157 Shed: 49
## Mean : 15.06 Mean : 2.759 NAs:1453 MnWw : 11 TenC: 1
## 3rd Qu.: 0.00 3rd Qu.: 0.000 NAs :1179 NAs :1406
## Max. :480.00 Max. :738.000
##
## MiscVal MoSold YrSold SaleType
## Min. : 0.00 Min. : 1.000 Min. :2006 WD :1267
## 1st Qu.: 0.00 1st Qu.: 5.000 1st Qu.:2007 New : 122
## Median : 0.00 Median : 6.000 Median :2008 COD : 43
## Mean : 43.49 Mean : 6.322 Mean :2008 ConLD : 9
## 3rd Qu.: 0.00 3rd Qu.: 8.000 3rd Qu.:2009 ConLI : 5
## Max. :15500.00 Max. :12.000 Max. :2010 ConLw : 5
## (Other): 9
## SaleCondition SalePrice
## Abnorml: 101 Min. : 34900
## AdjLand: 4 1st Qu.:129975
## Alloca : 12 Median :163000
## Family : 20 Mean :180921
## Normal :1198 3rd Qu.:214000
## Partial: 125 Max. :755000
##
```

Now that our data only includes numerical variables (for model simplicity purposes and curiosity of the power of this subset), we can set up our first model.

```
model <- lm(SalePrice ~ ., data = num_vars_df)
summary(model)
```

```
##
## Call:
## lm(formula = SalePrice ~ ., data = num_vars_df)
##
## Residuals:
## Min 1Q Median 3Q Max
## -497980 -16701 -1933 13823 308740
##
## Coefficients: (2 not defined because of singularities)
## Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.560e+05 1.431e+06 0.109 0.913252
## Id -1.291e+00 2.217e+00 -0.583 0.560288
## LotFrontage 5.264e+01 2.860e+01 1.840 0.065920 .
## LotArea 4.499e-01 1.019e-01 4.416 1.08e-05 ***
## OverallQual 1.647e+04 1.200e+03 13.725 < 2e-16 ***
## OverallCond 4.806e+03 1.037e+03 4.633 3.94e-06 ***
## YearBuilt 3.151e+02 6.174e+01 5.104 3.77e-07 ***
## YearRemodAdd 1.607e+02 6.700e+01 2.399 0.016587 *
```

```
## MasVnrArea      2.909e+01  6.020e+00  4.832 1.50e-06 ***
## BsmtFinSF1      2.010e+01  4.733e+00  4.248 2.30e-05 ***
## BsmtFinSF2      8.953e+00  7.155e+00  1.251 0.211016
## BsmtUnfSF       1.084e+01  4.253e+00  2.547 0.010957 *
## TotalBsmtSF      NA         NA         NA         NA
## X1stFlrSF       4.931e+01  5.850e+00  8.429 < 2e-16 ***
## X2ndFlrSF       4.147e+01  4.920e+00  8.428 < 2e-16 ***
## LowQualFinSF    1.789e+01  1.995e+01  0.897 0.369936
## GrLivArea        NA         NA         NA         NA
## BsmtFullBath     7.942e+03  2.638e+03  3.011 0.002652 **
## BsmtHalfBath     3.683e+02  4.140e+03  0.089 0.929118
## FullBath         3.151e+03  2.852e+03  1.105 0.269363
## HalfBath        -1.386e+03  2.702e+03 -0.513 0.608147
## BedroomAbvGr    -9.036e+03  1.704e+03 -5.304 1.31e-07 ***
## KitchenAbvGr    -2.447e+04  4.936e+03 -4.957 8.01e-07 ***
## TotRmsAbvGrd     5.860e+03  1.250e+03  4.688 3.02e-06 ***
## Fireplaces       3.637e+03  1.796e+03  2.025 0.043074 *
## GarageCars       1.048e+04  2.895e+03  3.621 0.000304 ***
## GarageArea       2.595e+00  9.832e+00  0.264 0.791851
## WoodDeckSF       2.601e+01  8.109e+00  3.207 0.001369 **
## OpenPorchSF      1.029e+00  1.537e+01  0.067 0.946635
## EnclosedPorch    1.328e+01  1.710e+01  0.777 0.437504
## X3SsnPorch       2.286e+01  3.184e+01  0.718 0.472869
## ScreenPorch      5.470e+01  1.743e+01  3.139 0.001731 **
## PoolArea        -4.008e+01  2.407e+01 -1.666 0.096014 .
## MiscVal         -1.977e-01  1.885e+00 -0.105 0.916476
## MoSold           2.957e+01  3.496e+02  0.085 0.932610
## YrSold           -5.777e+02  7.115e+02 -0.812 0.417022
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 35250 on 1426 degrees of freedom
## Multiple R-squared:  0.8076, Adjusted R-squared:  0.8031
## F-statistic: 181.3 on 33 and 1426 DF, p-value: < 2.2e-16
```

There are a lot of statistically insignificant variables which we will remove one by one or in sets if the p-value is very large, as they will never be significant, so anything above 0.5 will be removed right away and also log SalePrice as this is common practice given the exponential nature of house prices.

```
model11 <- lm(SalePrice ~ ., data = num_vars_df[, -c(1,12,16)] )
#summary(model11)
model12 <- lm(SalePrice ~ ., data = num_vars_df[, -c(1,12,16,33,34,35,28,18)] )
#summary(model12)
model13 <- lm(SalePrice ~ ., data = num_vars_df[, -c(1,12,16,33,34,35,28,18,26,20)] )
#summary(model13)
model14 <- lm(SalePrice ~ ., data = num_vars_df[, -c(1,12,16,33,34,35,28,18,26,20,30,29,15)] )
#summary(model14)
model15 <- lm(log(SalePrice) ~ ., data = num_vars_df[, -c(1,12,16,33,34,35,28,18,26,20,30,29,15)] )
#summary(model15)
model16 <- lm(log(SalePrice) ~ ., data = num_vars_df[, -c(1,12,16,33,34,35,28,18,26,20,30,29,15,2,8,21)] )
summary(model16)
```

```
##
```

```
## Call:
## lm(formula = log(SalePrice) ~ ., data = num_vars_df[, -c(1, 12,
##      16, 33, 34, 35, 28, 18, 26, 20, 30, 29, 15, 2, 8, 21)])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.01055 -0.06767  0.00578  0.07979  0.51970
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.048e+00  5.259e-01   5.796 8.35e-09 ***
## LotArea      2.035e-06  4.274e-07   4.761 2.12e-06 ***
## OverallQual  8.218e-02  4.910e-03  16.737 < 2e-16 ***
## OverallCond  4.913e-02  4.275e-03  11.492 < 2e-16 ***
## YearBuilt    2.746e-03  2.262e-04  12.139 < 2e-16 ***
## YearRemodAdd 1.069e-03  2.750e-04   3.887 0.000106 ***
## BsmtFinSF1   8.774e-05  1.947e-05   4.507 7.09e-06 ***
## BsmtFinSF2   7.940e-05  2.984e-05   2.661 0.007889 **
## BsmtUnfSF    6.262e-05  1.772e-05   3.534 0.000422 ***
## X1stFlrSF    2.054e-04  2.415e-05   8.506 < 2e-16 ***
## X2ndFlrSF    1.660e-04  1.754e-05   9.465 < 2e-16 ***
## BsmtFullBath 5.429e-02  1.054e-02   5.149 2.98e-07 ***
## FullBath     2.731e-02  1.093e-02   2.498 0.012617 *
## KitchenAbvGr -1.028e-01  2.052e-02  -5.008 6.18e-07 ***
## TotRmsAbvGrd 2.028e-02  4.543e-03   4.463 8.70e-06 ***
## Fireplaces   4.503e-02  7.425e-03   6.065 1.69e-09 ***
## GarageCars   7.724e-02  7.179e-03  10.758 < 2e-16 ***
## WoodDeckSF   1.107e-04  3.374e-05   3.280 0.001061 **
## ScreenPorch  3.371e-04  7.246e-05   4.653 3.58e-06 ***
## PoolArea     -3.674e-04  9.977e-05  -3.682 0.000240 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1487 on 1440 degrees of freedom
## Multiple R-squared:  0.8632, Adjusted R-squared:  0.8614
## F-statistic: 478.2 on 19 and 1440 DF, p-value: < 2.2e-16
```

We now have a model with statistically significant variables that explains 86.32% of the variation in logged house sale prices (Adjusted  $R^2 = 0.8632$ ). This represents a strong linear model. However, before proceeding with interpretation or submission, we must verify that the model satisfies the key assumptions of linear regression. Only then can we be confident that the results are reliable and statistically valid.

We must verify that the model satisfies the key assumptions of linear regression:

- **Linearity:** The relationship between the predictors and the response variable is linear.
- **Independence of errors:** The residuals are independent of each other.
- **Homoscedasticity:** The variance of the residuals is constant across all levels of the fitted values.
- **Normality of residuals:** The residuals are approximately normally distributed.
- **No severe multicollinearity:** The predictors are not highly correlated with each other.

In the following section, we perform a series of diagnostic checks to assess whether these assumptions hold.

```
library(tidyverse)
library(performance) # for nice checks
library(car)          # for VIF and other diagnostics
library(ggfortify)    # optional: ggplot2 versions of base plots
library(patchwork)    # for combining plots nicely
```

```
## Ignoring unknown labels:
## * size : ""
```



In the Posterior Predictive Checks section of the above plot, we can see that the model fits well for most regions of SalesPrice, except for around the median value of Sales price. This pairs well with the Linearity plot, as it is clear that the reference line is not linear and the points also deviate from the line, hence we should check for some non-linear relationships between our predictors and Sales price.

There is also evidence to suggest that there is non-equal variance (non-homogeneity of variance) as the reference line is clearly not flat, showing a quadratic shape, although the spread above and below the line is quite uniform, which is good.

There appears to be an influential observation as shown in the Cook's Distance Contour plot, observation 299 should be removed as its residual standard deviation is too far from the expected deviation for us to keep it without it influencing our model negatively.

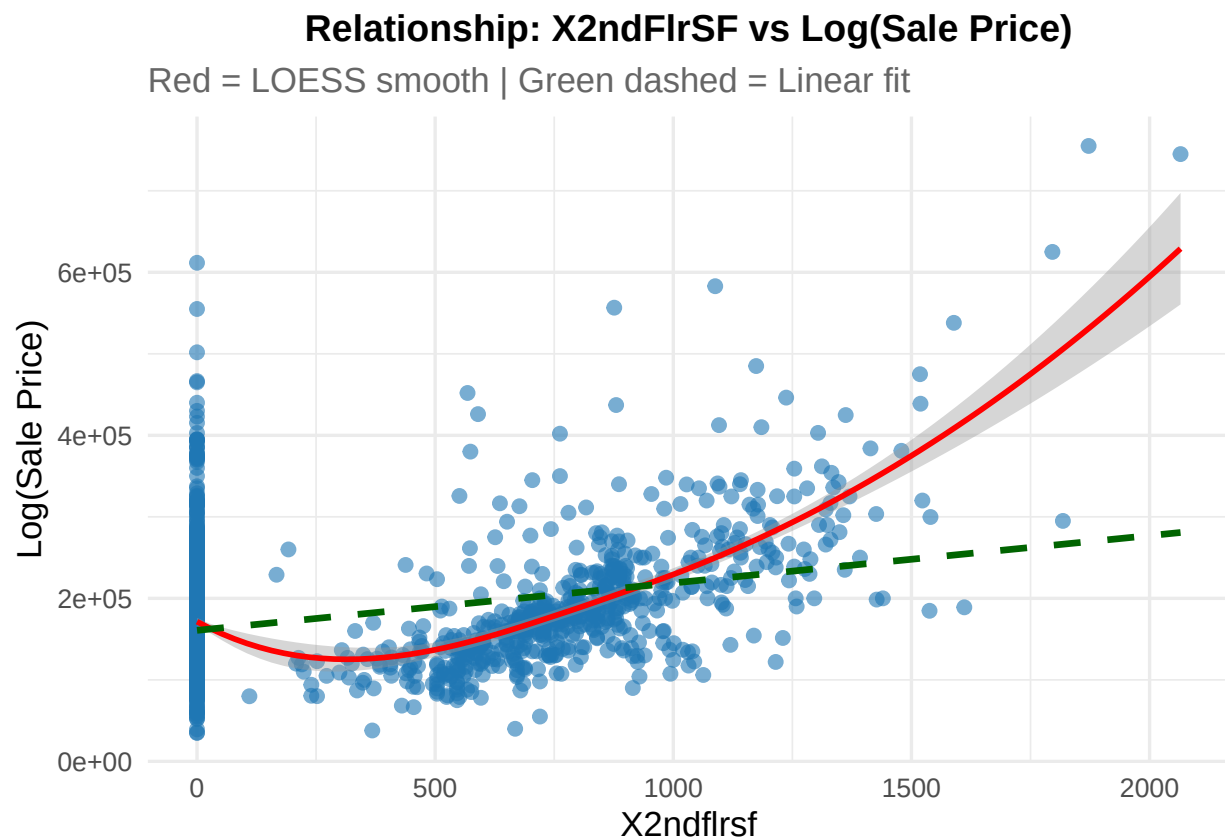
Some evidence of multicollinearity in our model, 5 of the variables with a variance inflation factor (VIF)  $\geq 5$ , hence we should consider which variables these are and remove the collinear variables.

And finally, a look at the Normality of the residuals shows us that they are fairly normal except for the tails where there is some clear deviation. Once we address the non-linear relationships, remove the influential observation and the collinear variables, this plot should follow the line more closely.

## Addressing Model Assumption Findings

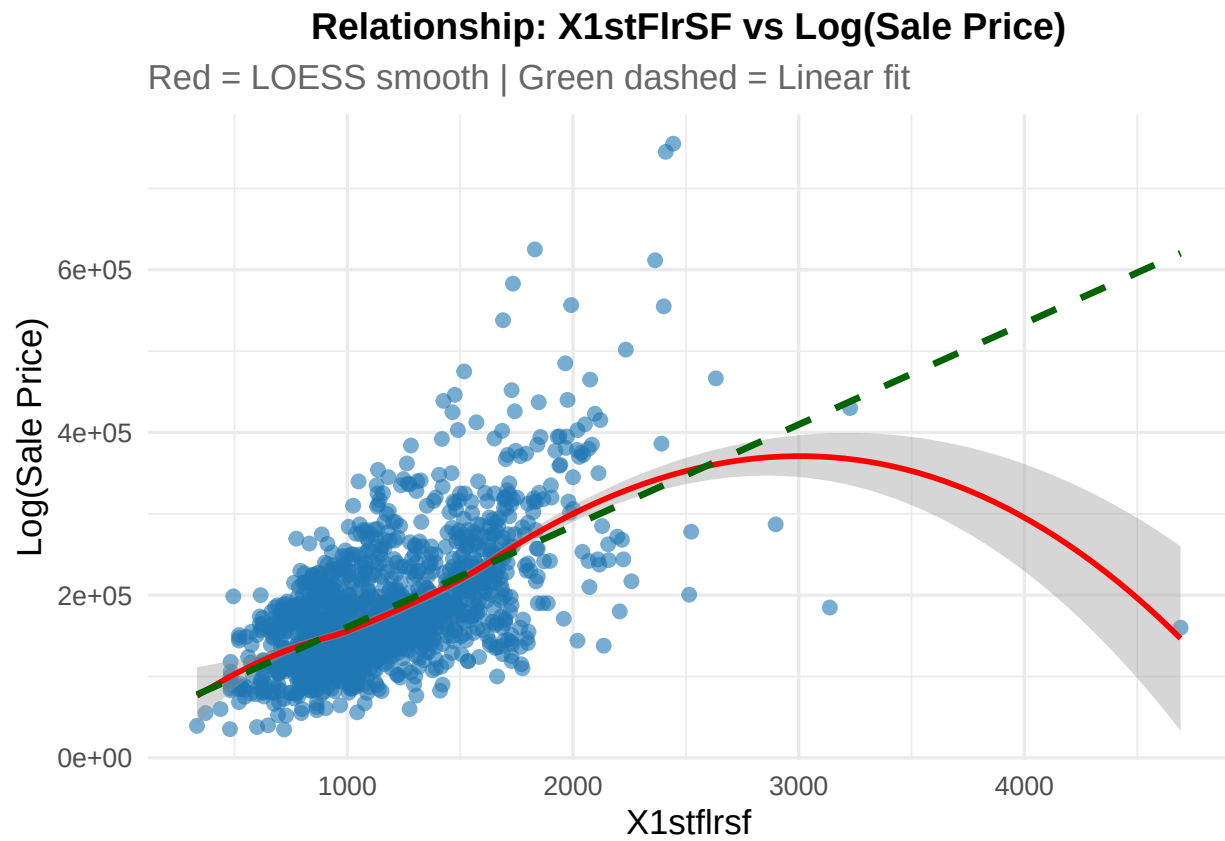
As there are a few too many variables to show all of the plot I will show the variables which shows a clear non-linear relationship with the log of sale price.

```
## [[1]]
```



```
##  
## [[2]]
```

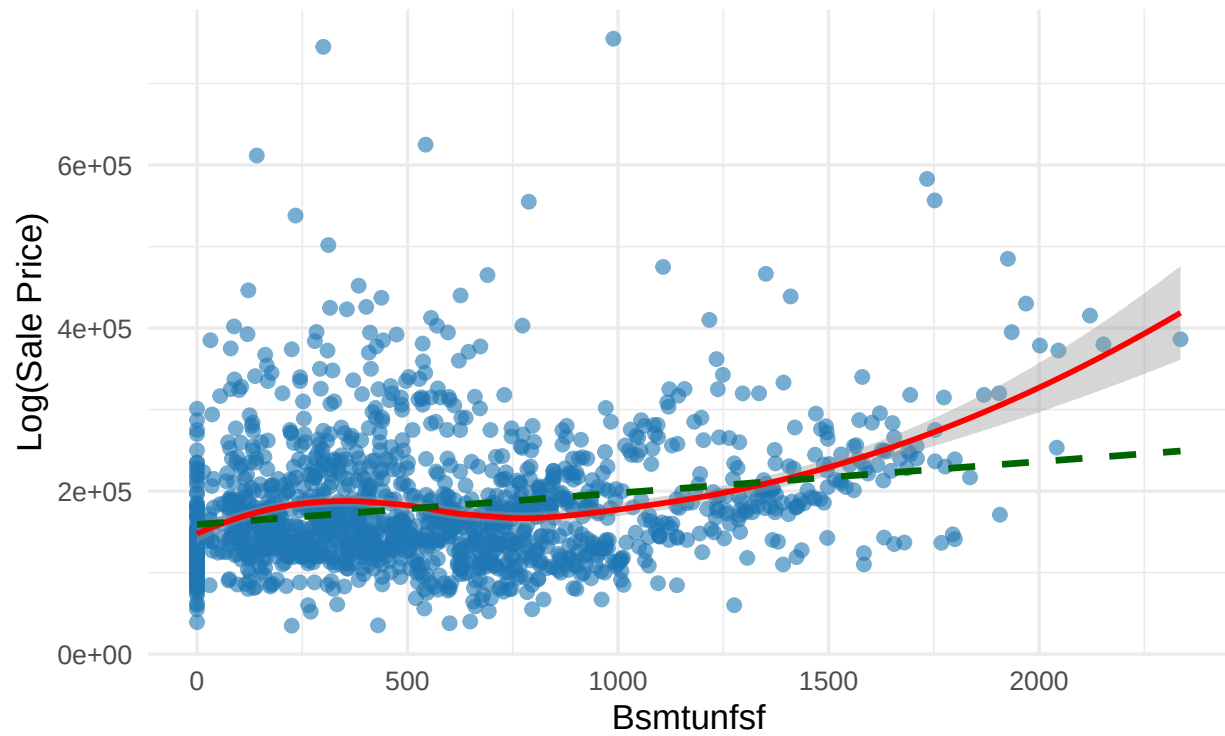




```
##  
## [[3]]
```

### Relationship: BsmtUnfSF vs Log(Sale Price)

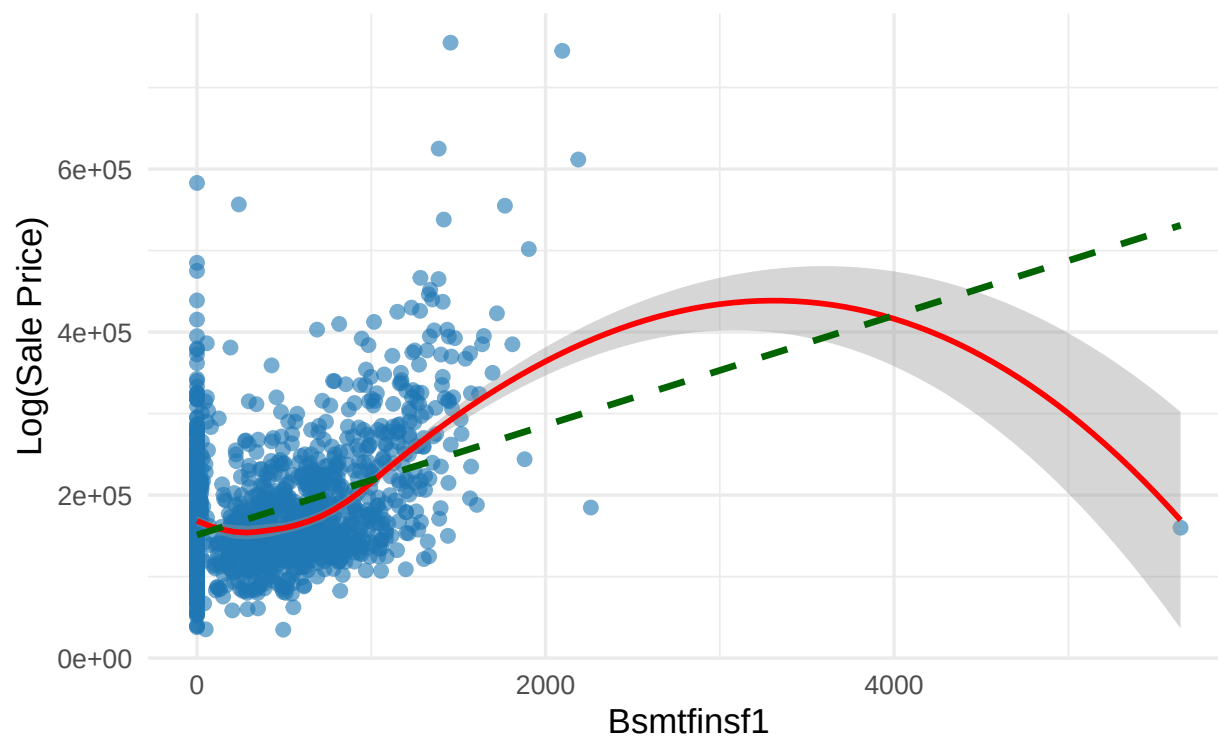
Red = LOESS smooth | Green dashed = Linear fit



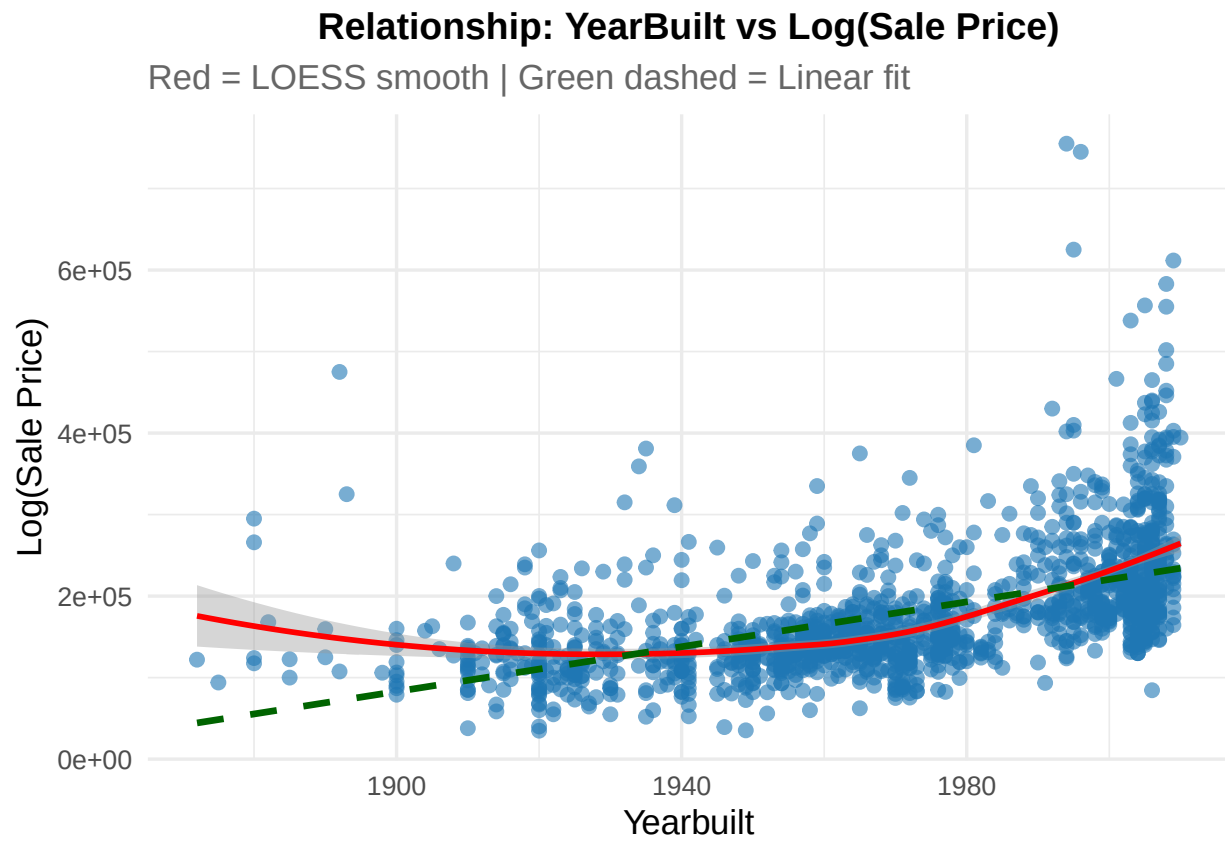
```
##  
## [[4]]
```

### Relationship: BsmtFinSF1 vs Log(Sale Price)

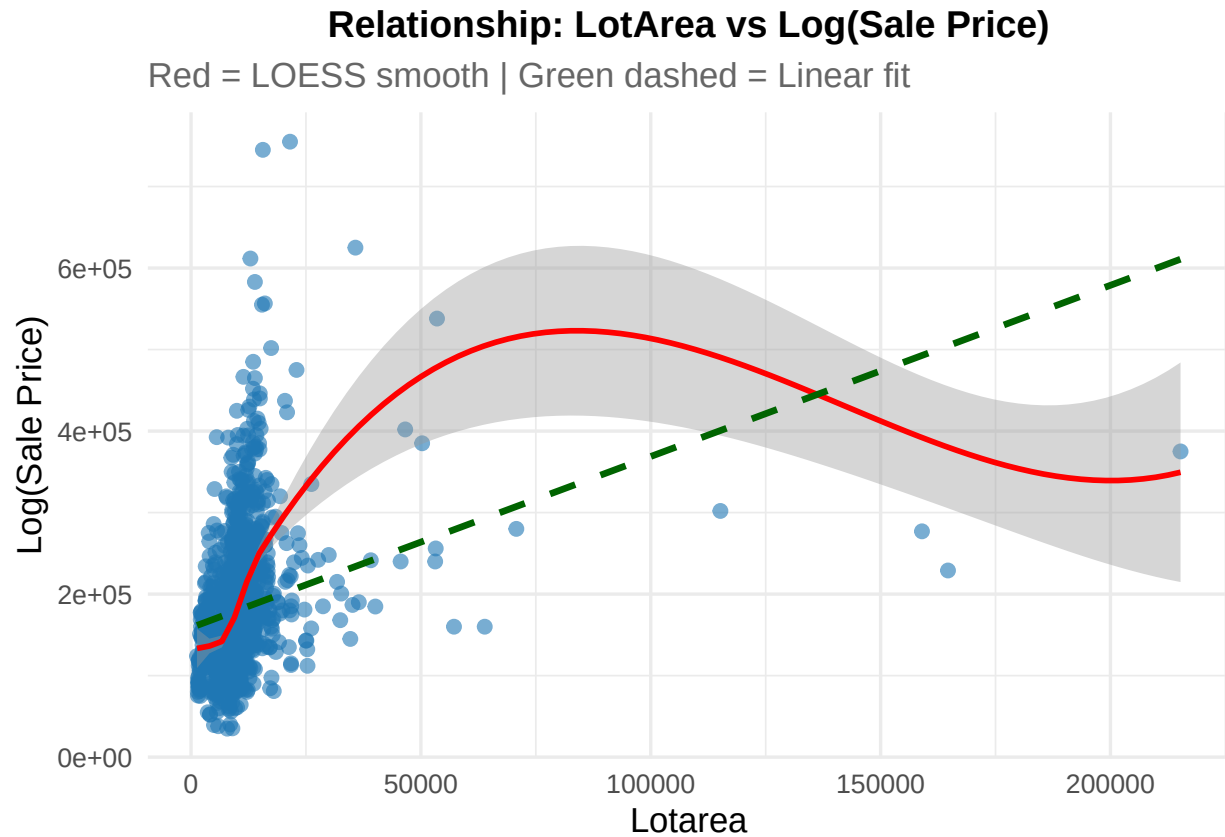
Red = LOESS smooth | Green dashed = Linear fit



```
##  
## [[5]]
```



```
##  
## [[6]]
```

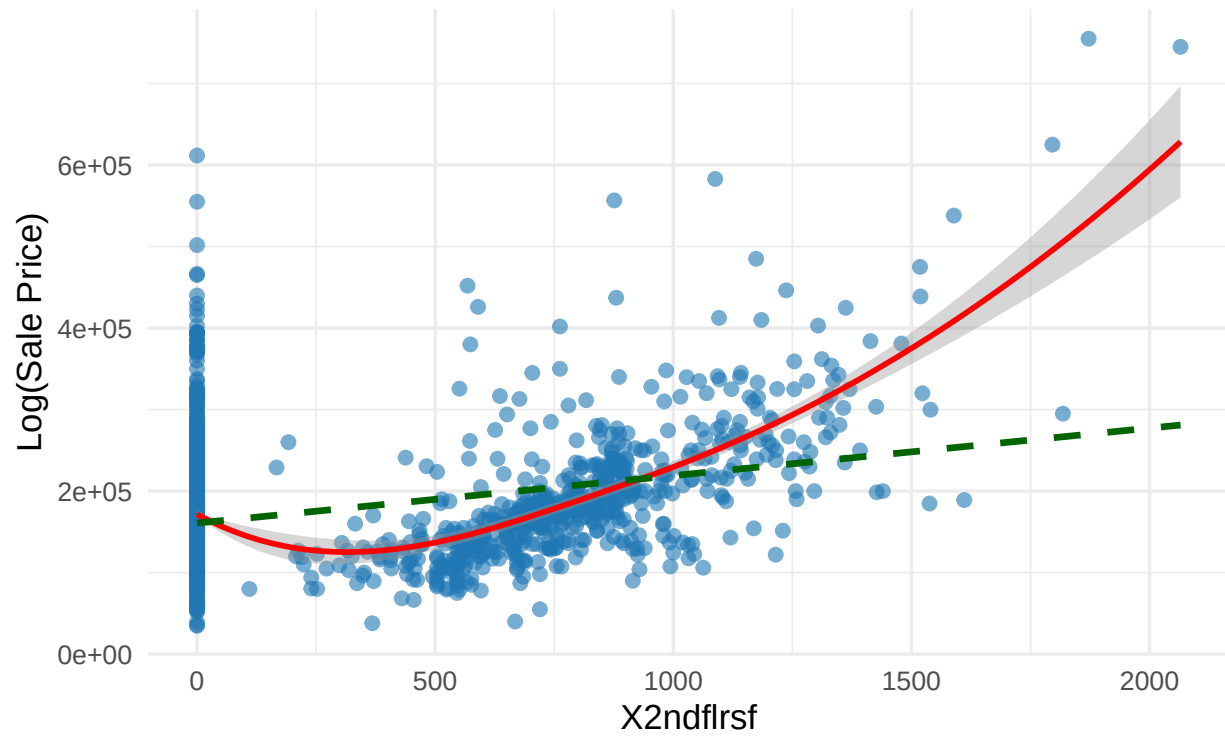


As we can see, there are some clearly non-linear relationships that need to be taken care of to improve our model and meet the required assumptions of a linear model. But a few of the plots show a non-linear relationship due to one observation dragging the line away, hence we should take care to remove this observation and check for linearity again.

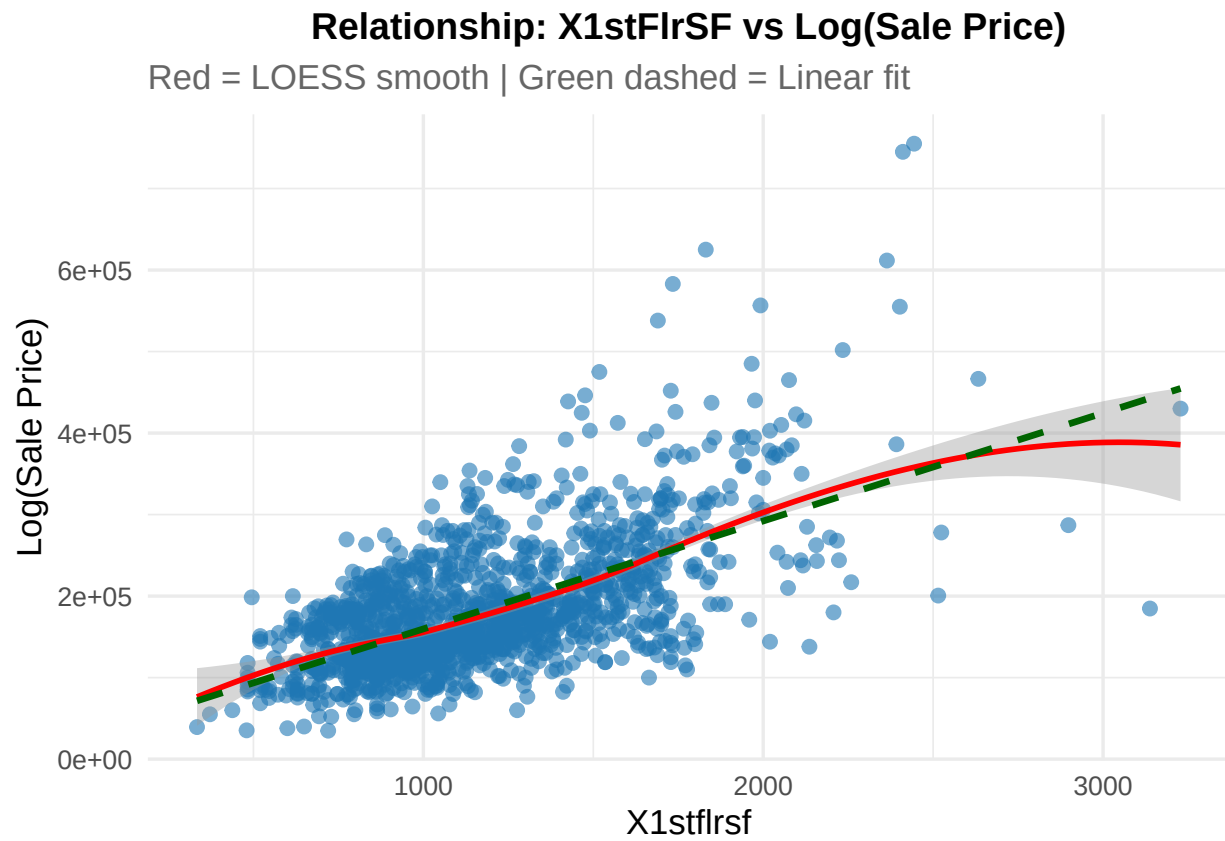
```
## [[1]]
```

### Relationship: X2ndFlrSF vs Log(Sale Price)

Red = LOESS smooth | Green dashed = Linear fit



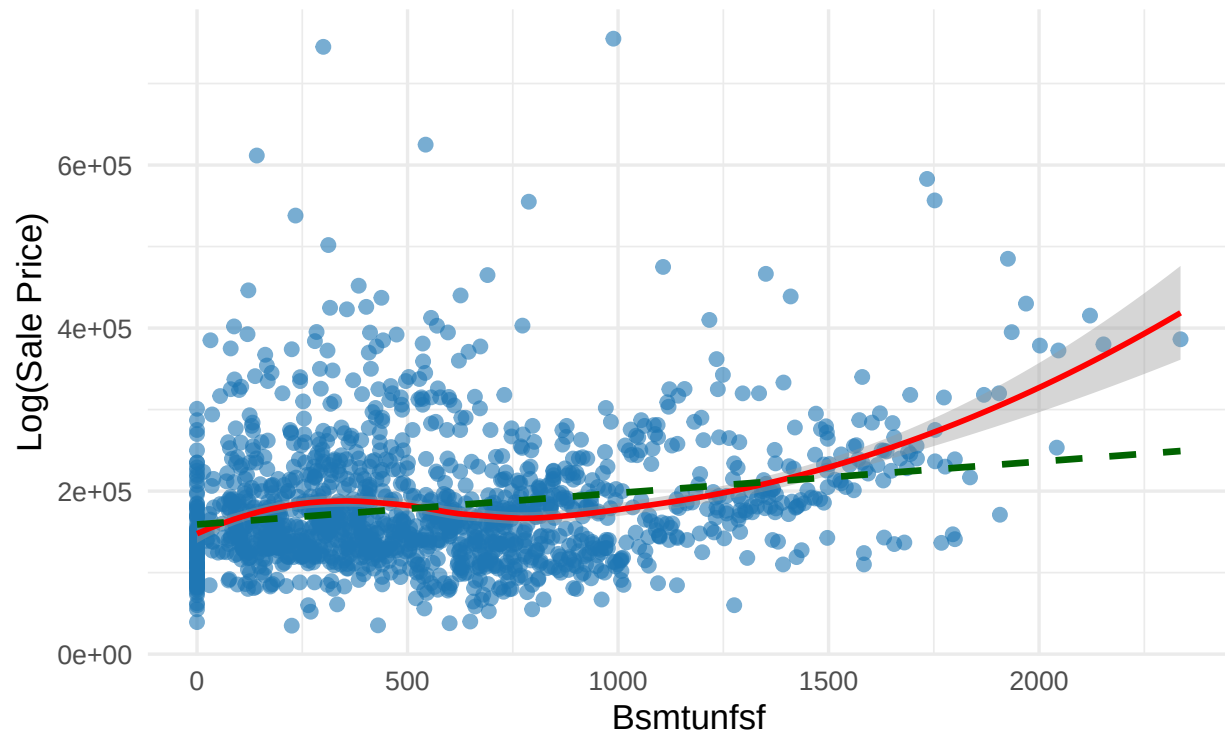
```
##  
## [[2]]
```



```
##  
## [[3]]
```

### Relationship: BsmtUnfSF vs Log(Sale Price)

Red = LOESS smooth | Green dashed = Linear fit

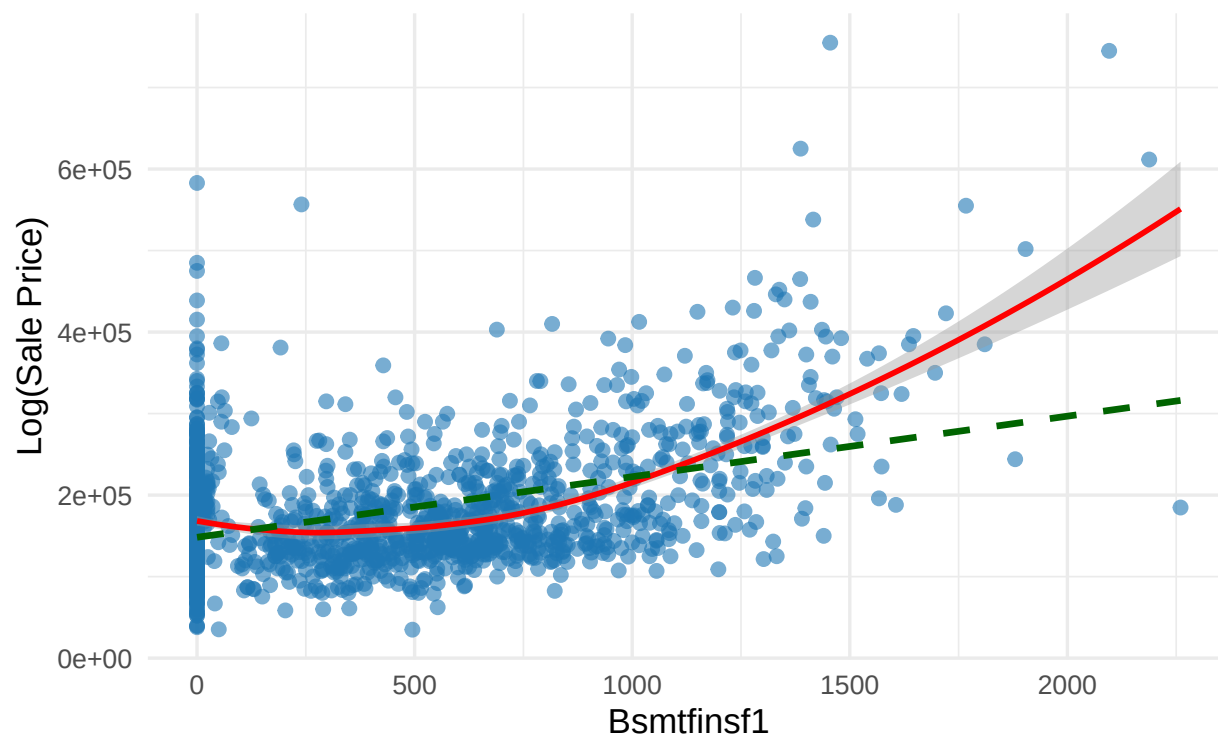


```
##  
## [[4]]
```

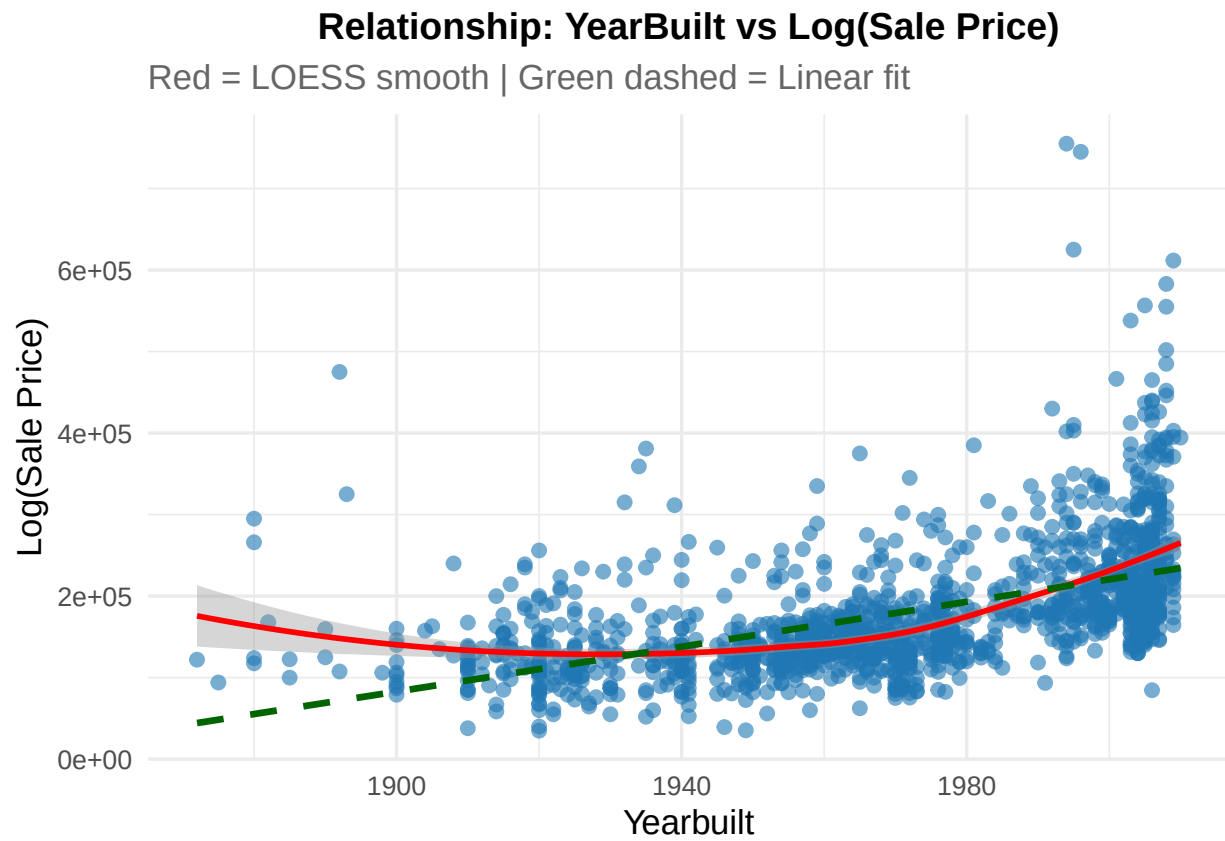


## Relationship: BsmtFinSF1 vs Log(Sale Price)

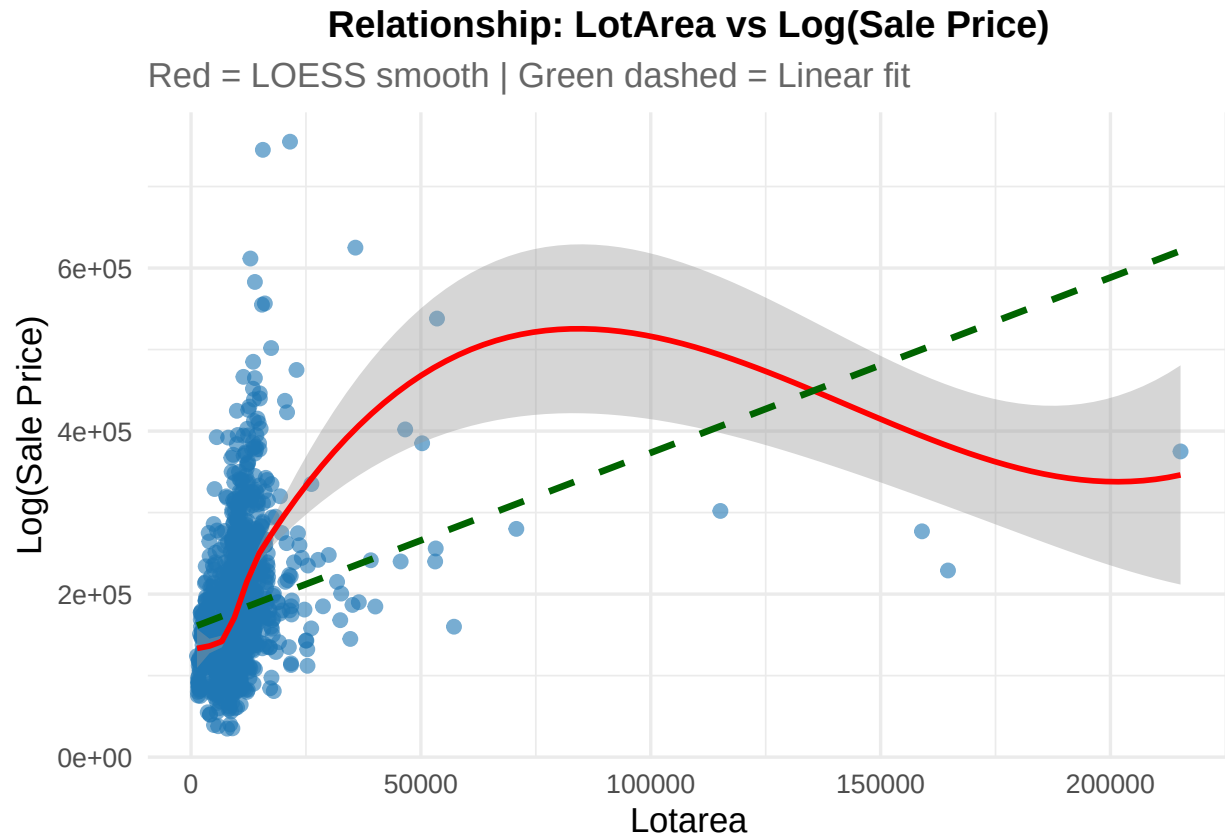
Red = LOESS smooth | Green dashed = Linear fit



```
##  
## [[5]]
```



```
##  
## [[6]]
```



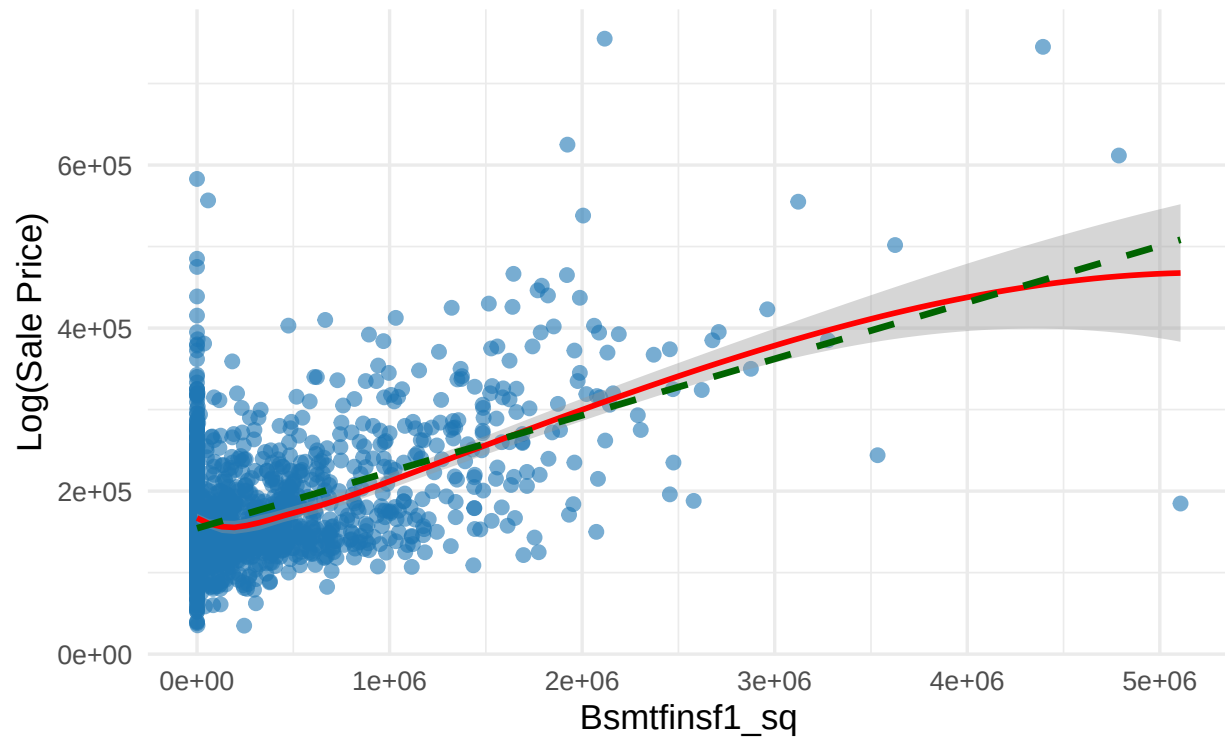
Removing the influential observation, was a wise choice, it was changing the relationship drastically. Regardless, these variables still have a nonlinear relationship which should be taken care of.

LotArea should definitely be logged as some houses have over 100000 square foot lot which influence the relationship greatly and we will put a quadratic on all other variables here except for X1.

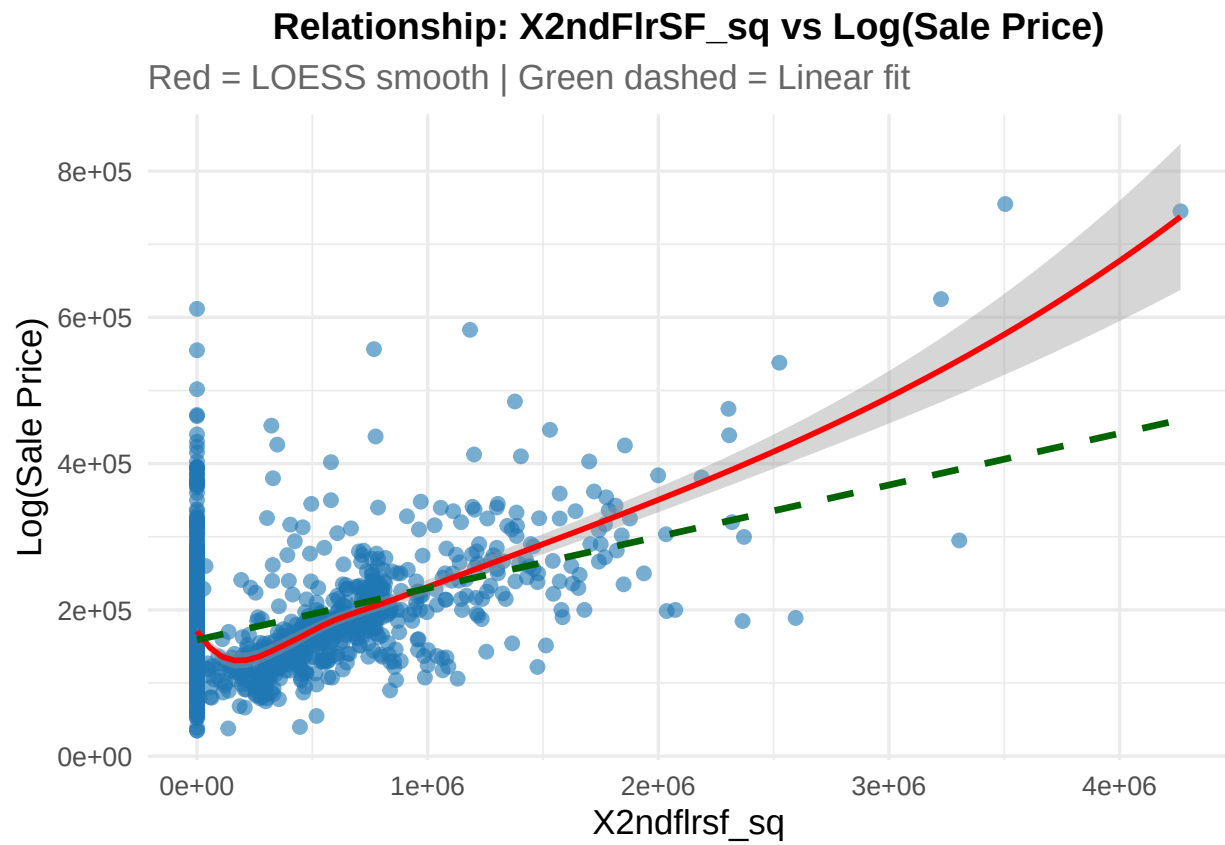
```
## [[1]]
```

### Relationship: BsmtFinSF1\_sq vs Log(Sale Price)

Red = LOESS smooth | Green dashed = Linear fit



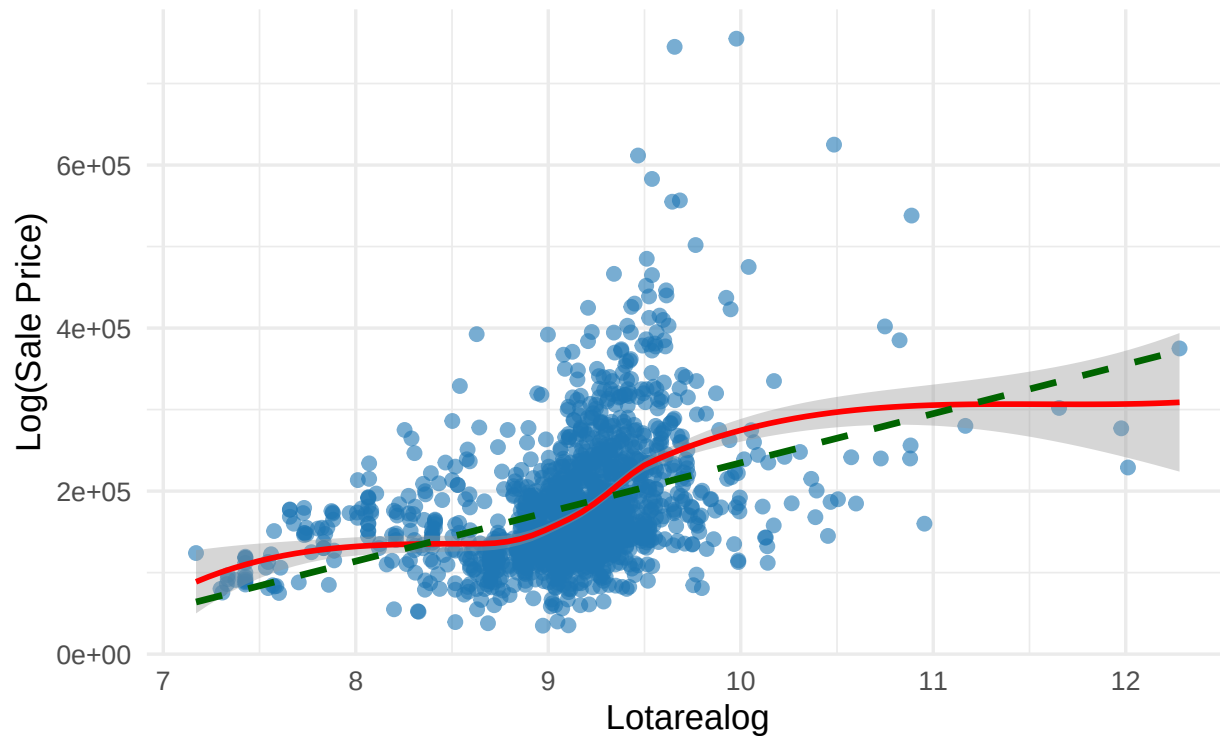
```
##  
## [[2]]
```



```
##  
## [[3]]
```

## Relationship: LotAreaLog vs Log(Sale Price)

Red = LOESS smooth | Green dashed = Linear fit



This is definitely an improvement, however it appears some observations are still making the relationships askew, but we will check if they are influencing the model afterwards. Now we will consider multicollinearity.

*#more details*

```
collinearity <- check_collinearity(model)
print(collinearity)
```

```
## # Check for Multicollinearity
```

```
##
```

```
## Low Correlation
```

```
##
```

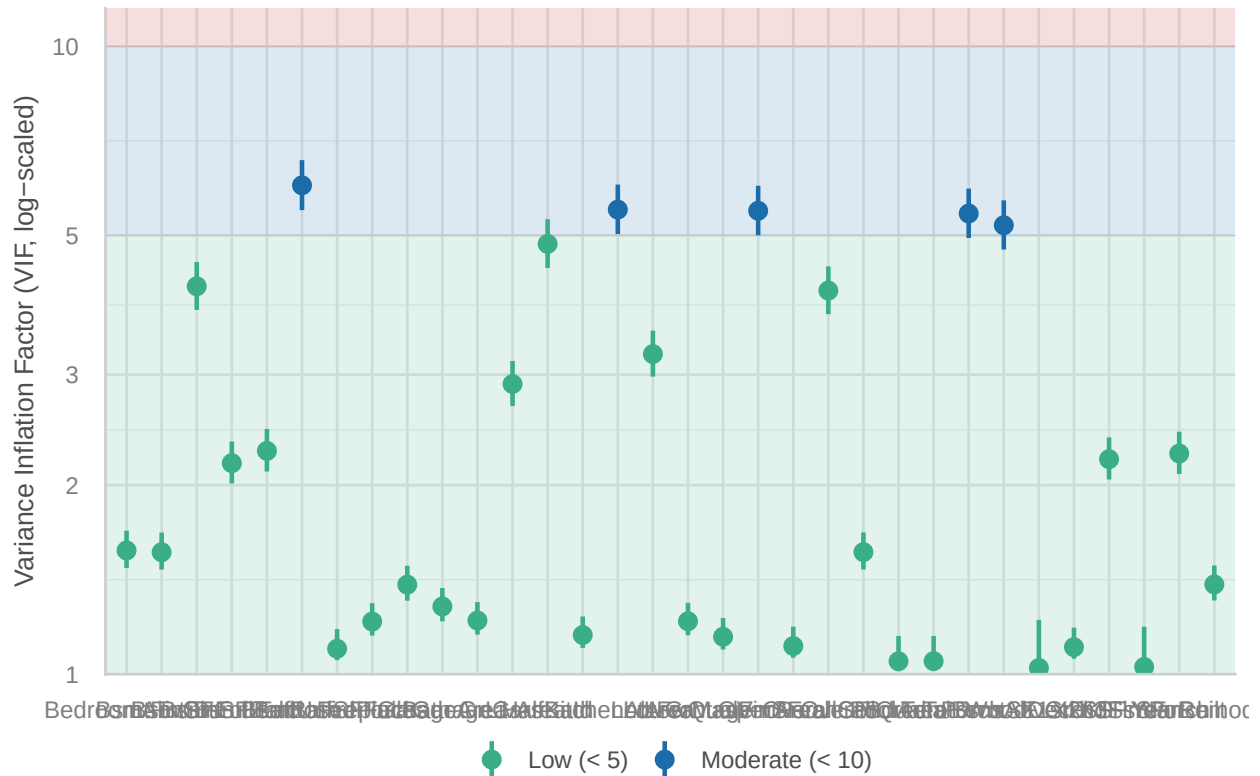
|    | Term         | VIF  | VIF 95% CI   | adj. VIF | Tolerance | Tolerance 95% CI |
|----|--------------|------|--------------|----------|-----------|------------------|
| ## | Id           | 1.15 | [1.10, 1.24] | 1.07     | 0.87      | [0.81, 0.91]     |
| ## | LotFrontage  | 1.21 | [1.15, 1.30] | 1.10     | 0.82      | [0.77, 0.87]     |
| ## | LotArea      | 3.24 | [2.98, 3.53] | 1.80     | 0.31      | [0.28, 0.34]     |
| ## | OverallQual  | 1.56 | [1.47, 1.68] | 1.25     | 0.64      | [0.59, 0.68]     |
| ## | OverallCond  | 4.08 | [3.74, 4.46] | 2.02     | 0.24      | [0.22, 0.27]     |
| ## | YearBuilt    | 2.25 | [2.08, 2.43] | 1.50     | 0.45      | [0.41, 0.48]     |
| ## | YearRemodAdd | 1.39 | [1.31, 1.49] | 1.18     | 0.72      | [0.67, 0.76]     |
| ## | BsmtFinSF1   | 1.56 | [1.47, 1.68] | 1.25     | 0.64      | [0.59, 0.68]     |
| ## | BsmtFinSF2   | 4.15 | [3.80, 4.53] | 2.04     | 0.24      | [0.22, 0.26]     |
| ## | X1stFlrSF    | 1.10 | [1.06, 1.19] | 1.05     | 0.91      | [0.84, 0.94]     |
| ## | X2ndFlrSF    | 2.20 | [2.04, 2.38] | 1.48     | 0.45      | [0.42, 0.49]     |
| ## | LowQualFinSF | 1.15 | [1.09, 1.23] | 1.07     | 0.87      | [0.81, 0.91]     |
| ## | GrLivArea    | 2.90 | [2.67, 3.15] | 1.70     | 0.34      | [0.32, 0.37]     |
| ## | BsmtFullBath | 2.17 | [2.01, 2.35] | 1.47     | 0.46      | [0.43, 0.50]     |
| ## | BsmtHalfBath | 2.27 | [2.10, 2.46] | 1.51     | 0.44      | [0.41, 0.48]     |

```
##      FullBath 1.39 [1.31, 1.49]      1.18      0.72      [0.67, 0.76]
##      HalfBath 4.85 [4.44, 5.31]      2.20      0.21      [0.19, 0.23]
##      BedroomAbvGr 1.57 [1.48, 1.69]      1.25      0.64      [0.59, 0.68]
##      Fireplaces 1.21 [1.15, 1.30]      1.10      0.82      [0.77, 0.87]
##      GarageCars 1.22 [1.16, 1.30]      1.10      0.82      [0.77, 0.86]
##      GarageArea 1.28 [1.21, 1.37]      1.13      0.78      [0.73, 0.82]
##      WoodDeckSF 1.02 [1.00, 1.22]      1.01      0.98      [0.82, 1.00]
##      OpenPorchSF 1.11 [1.06, 1.19]      1.05      0.90      [0.84, 0.94]
##      EnclosedPorch 1.10 [1.05, 1.18]      1.05      0.91      [0.85, 0.95]
##      X3SsnPorch 1.03 [1.00, 1.19]      1.01      0.97      [0.84, 1.00]
##      ScreenPorch 1.05 [1.02, 1.15]      1.02      0.95      [0.87, 0.98]
##      PoolArea 1.05 [1.02, 1.15]      1.02      0.95      [0.87, 0.98]
##
## Moderate Correlation
##
##      Term      VIF      VIF 95% CI      adj. VIF      Tolerance      Tolerance 95% CI
##      MasVnrArea 5.47 [5.00, 6.00]      2.34      0.18      [0.17, 0.20]
##      BsmtUnfSF 6.01 [5.49, 6.59]      2.45      0.17      [0.15, 0.18]
##      TotalBsmtSF 5.42 [4.95, 5.94]      2.33      0.18      [0.17, 0.20]
##      KitchenAbvGr 5.50 [5.02, 6.02]      2.34      0.18      [0.17, 0.20]
##      TotRmsAbvGrd 5.19 [4.75, 5.68]      2.28      0.19      [0.18, 0.21]
```

```
# Nice plot
plot(collinearity)
```

## Collinearity

High collinearity (VIF) may inflate parameter uncertainty



We should now remove one of the terms which have a moderate correlation but is also the least likely to be important to the model. BsmtUnfSF is the total square foot area of unfinished basement, it has the highest

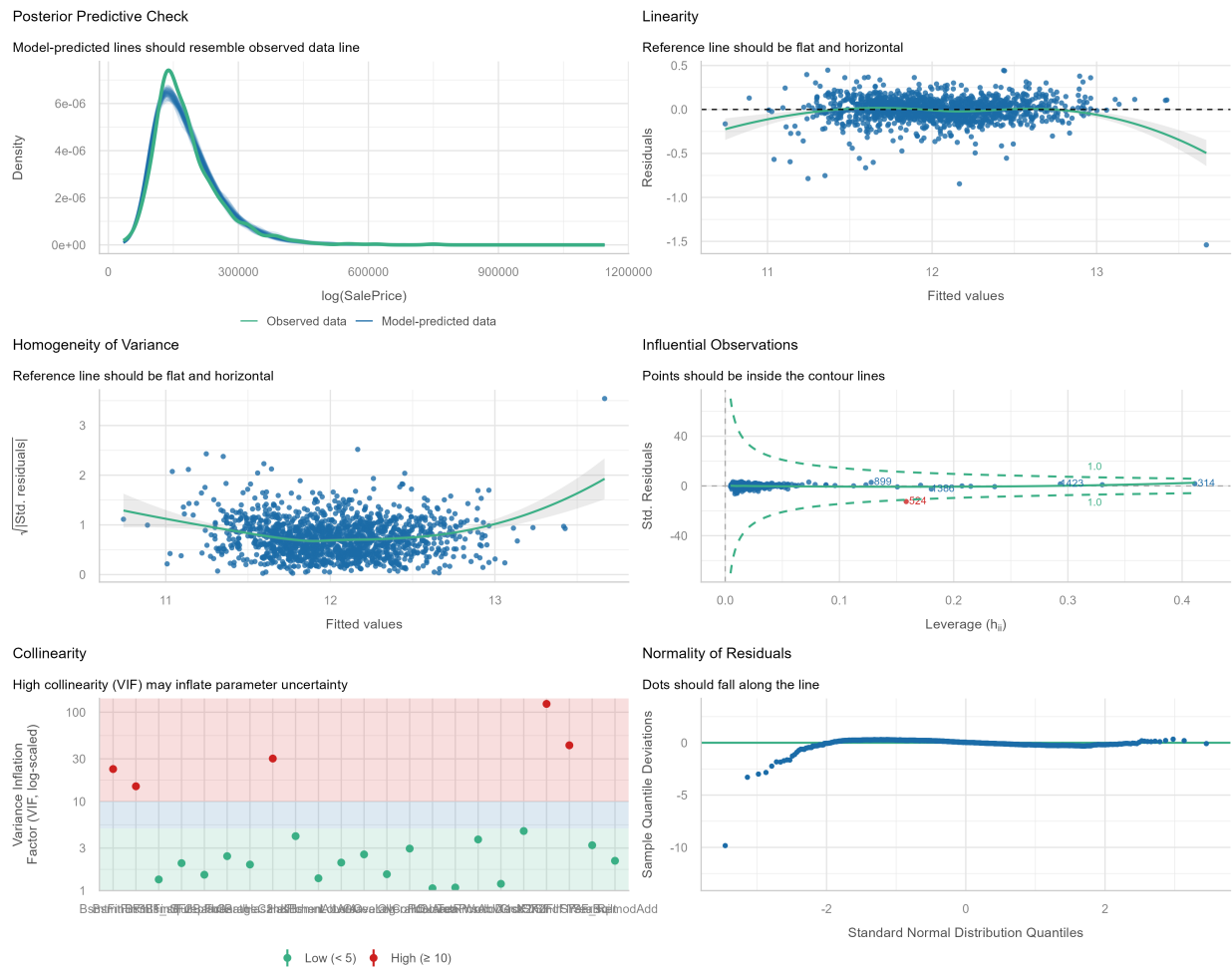
correlation and it may be useful but I will make the decision to remove it as this would perhaps only be important for very expensive houses who have unfinished work in the house.

```
modelnew <- lm(log(SalePrice) ~ ., data = num_vars_df[, -c(1, 11, 12, 16, 33, 34, 35, 28, 18, 26, 20, 30, 29, 15, 2, 8, 21)])
summary(modelnew)
```

```
##
## Call:
## lm(formula = log(SalePrice) ~ ., data = num_vars_df[, -c(1, 11,
##      12, 16, 33, 34, 35, 28, 18, 26, 20, 30, 29, 15, 2, 8, 21)])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.54005 -0.06190  0.00491  0.07221  0.44678
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.449e+00  5.002e-01   2.898  0.003816 **
## LotArea       -2.811e-07  5.094e-07  -0.552  0.581187
## OverallQual    8.751e-02  4.372e-03  20.015 < 2e-16 ***
## OverallCond    4.534e-02  3.893e-03  11.645 < 2e-16 ***
## YearBuilt     2.889e-03  2.086e-04  13.848 < 2e-16 ***
## YearRemodAdd  1.345e-03  2.496e-04   5.389 8.28e-08 ***
## BsmtFinSF1     1.019e-04  3.866e-05   2.635 0.008502 **
## BsmtFinSF2     2.193e-05  2.510e-05   0.874 0.382329
## X1stFlrSF      2.964e-04  2.015e-05  14.707 < 2e-16 ***
## X2ndFlrSF      3.428e-04  8.941e-05   3.834 0.000132 ***
## BsmtFullBath   2.898e-02  9.658e-03   3.001 0.002741 **
## FullBath       1.239e-02  9.928e-03   1.248 0.212237
## KitchenAbvGr  -9.166e-02  1.866e-02  -4.912 1.00e-06 ***
## TotRmsAbvGrd   7.957e-03  4.190e-03   1.899 0.057766 .
## Fireplaces     3.428e-02  6.708e-03   5.111 3.63e-07 ***
## GarageCars     5.788e-02  6.567e-03   8.814 < 2e-16 ***
## WoodDeckSF     6.603e-05  3.055e-05   2.161 0.030824 *
## ScreenPorch    2.474e-04  6.542e-05   3.782 0.000162 ***
## PoolArea       4.516e-05  9.466e-05   0.477 0.633382
## LotAreaLog     9.281e-02  1.086e-02   8.544 < 2e-16 ***
## Has2ndFlr     -5.536e-02  3.890e-02  -1.423 0.154944
## X2ndFlrSF_sq  -7.038e-08  4.689e-08  -1.501 0.133605
## HasBsmnt       8.406e-03  1.518e-02   0.554 0.579698
## BsmtFinSF1_sq -2.232e-08  2.330e-08  -0.958 0.338299
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1339 on 1435 degrees of freedom
## Multiple R-squared:  0.8896, Adjusted R-squared:  0.8878
## F-statistic: 502.5 on 23 and 1435 DF,  p-value: < 2.2e-16

## Ignoring unknown labels:
## * size : ""
```





Great improvement in the linear model assumptions, we can see that the model prediction is very close to the observed data across all ranges now but we have introduced some strong multicollinearity but this is due to the non-linear predictors we added to the model, but to confirm this, let's check it. We should also remove observation 524, as it clearly is outside of the range of expected values.

```
collinearity <- check_collinearity(modelnew)
print(collinearity)
```

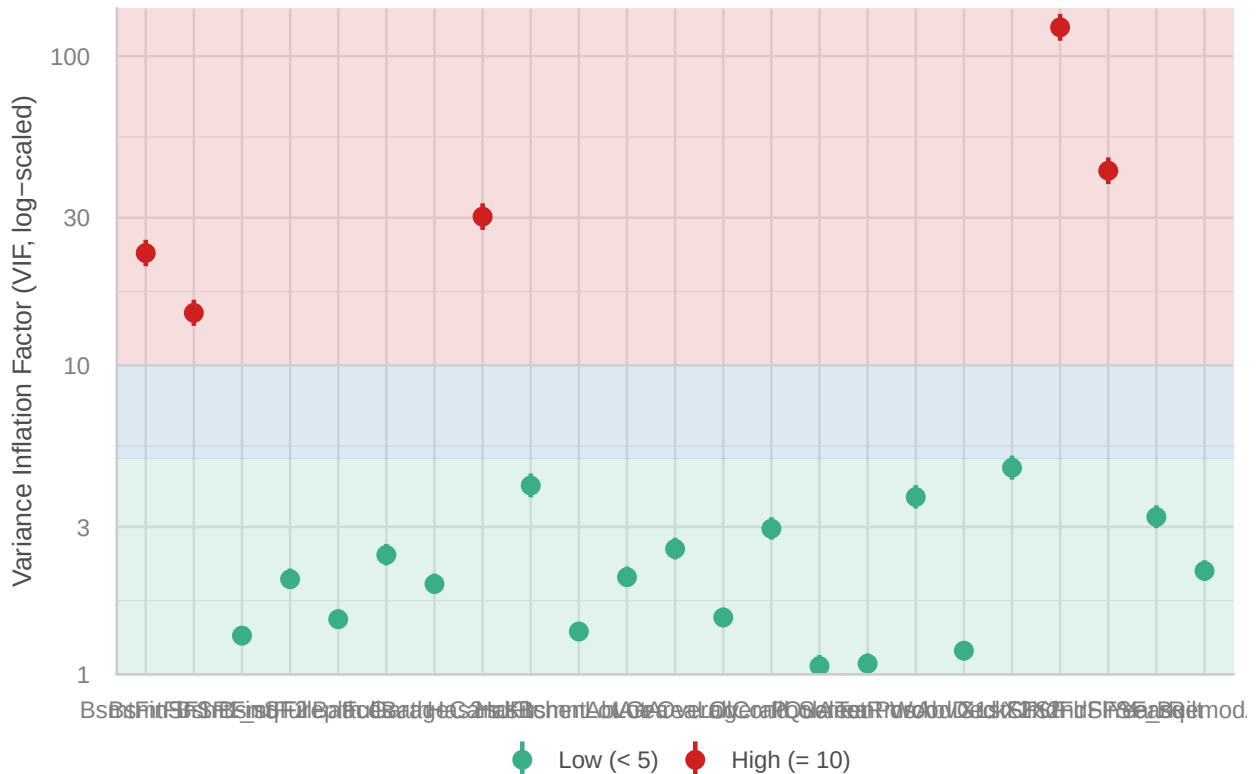
```
## # Check for Multicollinearity
##
## Low Correlation
##
##      Term  VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##      LotArea 2.06 [ 1.92,  2.23]   1.44    0.48 [0.45, 0.52]
##      OverallQual 2.96 [ 2.73,  3.23]   1.72    0.34 [0.31, 0.37]
##      OverallCond 1.53 [ 1.43,  1.64]   1.24    0.65 [0.61, 0.70]
##      YearBuilt 3.23 [ 2.97,  3.52]   1.80    0.31 [0.28, 0.34]
##      YearRemodAdd 2.16 [ 2.01,  2.34]   1.47    0.46 [0.43, 0.50]
##      BsmtFinSF2 1.33 [ 1.26,  1.43]   1.16    0.75 [0.70, 0.79]
##      X1stFlrSF 4.66 [ 4.26,  5.10]   2.16    0.21 [0.20, 0.23]
##      BsmtFullBath 2.03 [ 1.89,  2.20]   1.43    0.49 [0.45, 0.53]
##      FullBath 2.44 [ 2.25,  2.64]   1.56    0.41 [0.38, 0.44]
```

```
## KitchenAbvGr 1.38 [ 1.30, 1.48] 1.17 0.73 [0.68, 0.77]
## TotRmsAbvGrd 3.75 [ 3.44, 4.10] 1.94 0.27 [0.24, 0.29]
## Fireplaces 1.51 [ 1.42, 1.62] 1.23 0.66 [0.62, 0.71]
## GarageCars 1.96 [ 1.82, 2.12] 1.40 0.51 [0.47, 0.55]
## WoodDeckSF 1.19 [ 1.13, 1.28] 1.09 0.84 [0.78, 0.88]
## ScreenPorch 1.08 [ 1.04, 1.17] 1.04 0.92 [0.86, 0.96]
## PoolArea 1.06 [ 1.03, 1.16] 1.03 0.94 [0.87, 0.97]
## LotAreaLog 2.55 [ 2.35, 2.77] 1.60 0.39 [0.36, 0.42]
## HasBsmnt 4.08 [ 3.74, 4.46] 2.02 0.25 [0.22, 0.27]
##
## High Correlation
##
##      Term      VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
## BsmtFinSF1 23.07 [ 20.90, 25.46] 4.80 0.04 [0.04, 0.05]
## X2ndFlrSF 123.89 [112.06, 136.97] 11.13 8.07e-03 [0.01, 0.01]
## Has2ndFlr 30.24 [ 27.39, 33.40] 5.50 0.03 [0.03, 0.04]
## X2ndFlrSF_sq 42.56 [ 38.53, 47.02] 6.52 0.02 [0.02, 0.03]
## BsmtFinSF1_sq 14.77 [ 13.41, 16.29] 3.84 0.07 [0.06, 0.07]
```

```
# Nice plot
plot(collinearity)
```

## Collinearity

High collinearity (VIF) may inflate parameter uncertainty



As expected, the high correlation variables are the non-linear terms and the logical variables, we will not remove them for that reason.

```

#Removing the influential observation
#num_vars_df <- num_vars_df[-524,]

modelnew <- lm(log(SalePrice) ~ ., data = num_vars_df[, -c(1, 11, 12, 16, 33, 34, 35, 28, 18, 26, 20, 30, 29, 15, 2, 8, 21)],
summary(modelnew)

##
## Call:
## lm(formula = log(SalePrice) ~ ., data = num_vars_df[, -c(1, 11,
##      12, 16, 33, 34, 35, 28, 18, 26, 20, 30, 29, 15, 2, 8, 21)])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.54005 -0.06190  0.00491  0.07221  0.44678
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.449e+00  5.002e-01   2.898  0.003816 **
## LotArea      -2.811e-07  5.094e-07  -0.552  0.581187
## OverallQual   8.751e-02  4.372e-03  20.015 < 2e-16 ***
## OverallCond   4.534e-02  3.893e-03  11.645 < 2e-16 ***
## YearBuilt     2.889e-03  2.086e-04  13.848 < 2e-16 ***
## YearRemodAdd  1.345e-03  2.496e-04   5.389  8.28e-08 ***
## BsmtFinSF1    1.019e-04  3.866e-05   2.635  0.008502 **
## BsmtFinSF2    2.193e-05  2.510e-05   0.874  0.382329
## X1stFlrSF     2.964e-04  2.015e-05  14.707 < 2e-16 ***
## X2ndFlrSF     3.428e-04  8.941e-05   3.834  0.000132 ***
## BsmtFullBath   2.898e-02  9.658e-03   3.001  0.002741 **
## FullBath      1.239e-02  9.928e-03   1.248  0.212237
## KitchenAbvGr  -9.166e-02  1.866e-02  -4.912  1.00e-06 ***
## TotRmsAbvGrd   7.957e-03  4.190e-03   1.899  0.057766 .
## Fireplaces    3.428e-02  6.708e-03   5.111  3.63e-07 ***
## GarageCars     5.788e-02  6.567e-03   8.814 < 2e-16 ***
## WoodDeckSF     6.603e-05  3.055e-05   2.161  0.030824 *
## ScreenPorch    2.474e-04  6.542e-05   3.782  0.000162 ***
## PoolArea      4.516e-05  9.466e-05   0.477  0.633382
## LotAreaLog     9.281e-02  1.086e-02   8.544 < 2e-16 ***
## Has2ndFlr     -5.536e-02  3.890e-02  -1.423  0.154944
## X2ndFlrSF_sq  -7.038e-08  4.689e-08  -1.501  0.133605
## HasBsmnt       8.406e-03  1.518e-02   0.554  0.579698
## BsmtFinSF1_sq -2.232e-08  2.330e-08  -0.958  0.338299
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1339 on 1435 degrees of freedom
## Multiple R-squared:  0.8896, Adjusted R-squared:  0.8878
## F-statistic: 502.5 on 23 and 1435 DF, p-value: < 2.2e-16

```

Our model is getting better and better. Now we can explain 90.17% of the variation in logged Sale Prices however we have two insignificant variables which we will remove only BsmtFinSF2 as the other is the a member of the non-linear variables we created before.

```
modelnew2 <- lm(log(SalePrice) ~ ., data = num_vars_df[, -c(1, 10, 11, 12, 16, 33, 34, 35, 28, 18, 26, 20, 30, 29, 15, 2, 8, 21)])
summary(modelnew2)
```

```
##
## Call:
## lm(formula = log(SalePrice) ~ ., data = num_vars_df[, -c(1, 10,
##      11, 12, 16, 33, 34, 35, 28, 18, 26, 20, 30, 29, 15, 2, 8,
##      21)])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.54368 -0.06194  0.00443  0.07198  0.44729
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.468e+00  4.996e-01   2.939 0.003345 **
## LotArea      -2.521e-07  5.083e-07  -0.496 0.619932
## OverallQual    8.734e-02  4.367e-03  19.998 < 2e-16 ***
## OverallCond    4.549e-02  3.890e-03  11.695 < 2e-16 ***
## YearBuilt     2.890e-03  2.086e-04  13.856 < 2e-16 ***
## YearRemodAdd  1.333e-03  2.493e-04   5.349 1.03e-07 ***
## BsmtFinSF1     9.170e-05  3.686e-05   2.488 0.012974 *
## X1stFlrSF     2.985e-04  2.000e-05  14.929 < 2e-16 ***
## X2ndFlrSF     3.479e-04  8.921e-05   3.899 0.000101 ***
## BsmtFullBath   3.106e-02  9.358e-03   3.320 0.000924 ***
## FullBath       1.229e-02  9.927e-03   1.238 0.215854
## KitchenAbvGr  -9.218e-02  1.865e-02  -4.943 8.61e-07 ***
## TotRmsAbvGrd   7.801e-03  4.186e-03   1.864 0.062578 .
## Fireplaces     3.431e-02  6.707e-03   5.116 3.54e-07 ***
## GarageCars     5.780e-02  6.566e-03   8.802 < 2e-16 ***
## WoodDeckSF     6.803e-05  3.046e-05   2.233 0.025682 *
## ScreenPorch    2.521e-04  6.520e-05   3.867 0.000115 ***
## PoolArea       4.844e-05  9.458e-05   0.512 0.608615
## LotAreaLog     9.292e-02  1.086e-02   8.556 < 2e-16 ***
## Has2ndFlr     -5.757e-02  3.882e-02  -1.483 0.138272
## X2ndFlrSF_sq  -7.278e-08  4.681e-08  -1.555 0.120221
## HasBsmnt       1.300e-02  1.424e-02   0.913 0.361333
## BsmtFinSF1_sq -1.914e-08  2.301e-08  -0.832 0.405746
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1338 on 1436 degrees of freedom
## Multiple R-squared:  0.8895, Adjusted R-squared:  0.8878
## F-statistic: 525.4 on 22 and 1436 DF, p-value: < 2.2e-16
```

The Pool Area is now insignificant, so we will remove variables one by one until all are significant.

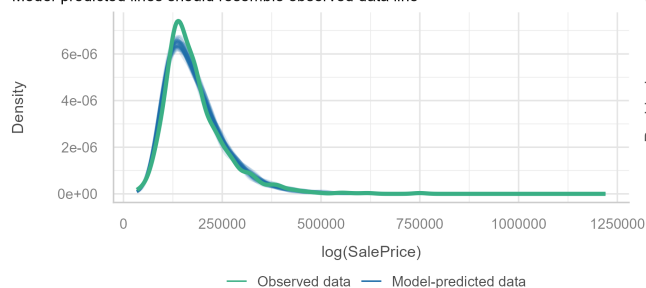
Now we have all significant variables in our model to the 10% level at least.

Lets check our assumptions:

```
## Ignoring unknown labels:
## * size : ""
```

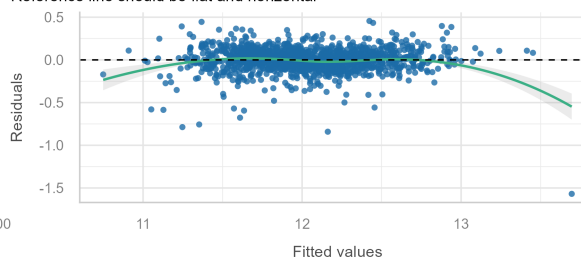
### Posterior Predictive Check

Model-predicted lines should resemble observed data line



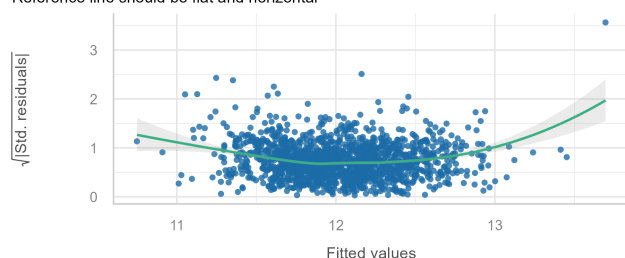
### Linearity

Reference line should be flat and horizontal



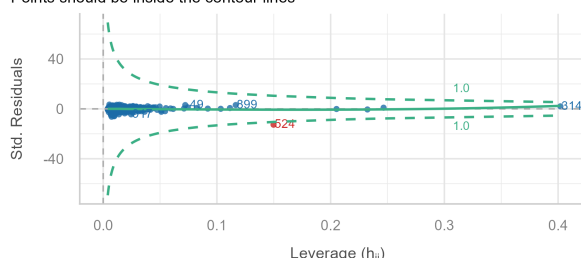
### Homogeneity of Variance

Reference line should be flat and horizontal



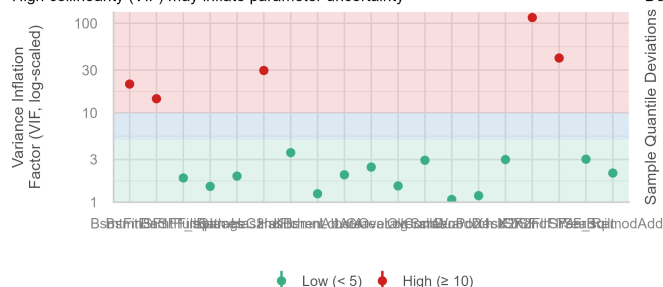
### Influential Observations

Points should be inside the contour lines



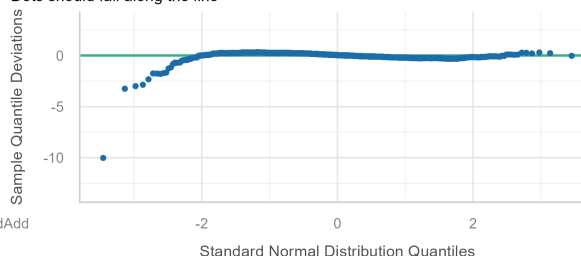
### Collinearity

High collinearity (VIF) may inflate parameter uncertainty



### Normality of Residuals

Dots should fall along the line



## Model Diagnostics Summary

Our final model satisfies the core assumptions of linear regression to a high standard:

- **Linearity:** The residuals versus fitted values plot shows a nearly random scatter around zero, indicating that the functional form (including polynomial terms where used) appropriately captures the relationships in the data.
- **Homoscedasticity:** The spread of residuals is largely consistent across fitted values, supporting the assumption of constant variance.
- **Normality of Residuals:** The residuals are approximately normally distributed around zero, with only a minor left skew. This small deviation is unlikely to meaningfully impact model predictions.
- **Multicollinearity:** After centering the polynomial terms, variance inflation factors (VIF) are within acceptable ranges.

Overall, the diagnostic checks confirm that the model is reliable and well-specified. With an Adjusted  $R^2$  of **90.16%**, the model explains a very high proportion of the variation in logged house sale prices.

We are now ready to generate predictions on the test dataset and submit our results to Kaggle.

From the logarithmic model of sales prices using only the numerical variables in the dataset we ended up with a RMSE of 0.15623, placing us in the top 60% only, hence we need to improve our model is not up to scratch.